# Magnetic Field: Concepts, Laws and Applications

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## 1. Introduction to Magnetic Fields

A **magnetic field** is a region of space around a magnet or a moving electric charge in which other magnets or moving charges experience a force. It is a vector field, which means that at every point in space it has both a magnitude and a direction. The magnetic field is represented by the symbol **B** and is also called magnetic flux density.

Magnetic effects have been known since ancient times, when people noticed that a naturally occurring mineral called lodestone attracted small pieces of iron. In 1820, Hans Christian Oersted discovered that an electric current deflects a nearby compass needle. This showed that electricity and magnetism are linked. Later, Michael Faraday and James Clerk Maxwell developed the theory of electromagnetism, which unified the two phenomena.

### 1.1 Units of magnetic field

In the SI system the unit of magnetic field is the **tesla (T)**. One tesla equals one newton per ampere-metre. A smaller unit, the **gauss (G)**, is also common, where 1 T = 10,000 G.

| Source | Approximate field strength |
|---|---|
| Earth's magnetic field at the surface | 25 to 65 microtesla |
| Small fridge magnet | about 5 millitesla |
| Loudspeaker magnet | about 1 tesla |
| Medical MRI scanner | 1.5 to 3 tesla |
| Strongest continuous lab magnets | over 40 tesla |

## 2. Sources of Magnetic Fields

### 2.1 Permanent magnets

In permanent magnets such as bar magnets, the magnetic field comes from the aligned magnetic moments of electrons inside the material. Every magnet has two poles, called north and south. Like poles repel each other and unlike poles attract. Magnetic poles always occur in pairs. If a bar magnet is cut in half, each half becomes a smaller magnet with its own north and south pole. No isolated magnetic monopole has ever been observed.

### 2.2 Moving charges and electric currents

Every moving charge produces a magnetic field. A steady electric current, which is a flow of many moving charges, therefore produces a magnetic field in the space around the conductor. This is the principle behind electromagnets, electric motors and transformers.

### 2.3 Magnetic field lines

Magnetic field lines are imaginary lines used to visualise the field. They follow a few simple rules:

- Field lines point from the north pole to the south pole outside the magnet, and from south to north inside it.
- Field lines form continuous closed loops. They never start or end anywhere.
- Field lines never cross each other, because the field has only one direction at each point.
- Where the lines are closer together, the field is stronger.

## 3. Magnetic Field of Current-Carrying Conductors

### 3.1 Straight wire

A long straight wire carrying a current *I* produces circular field lines around the wire. The direction is given by the **right-hand grip rule**: if the thumb of the right hand points along the current, the curled fingers show the direction of the magnetic field. The magnitude of the field at distance *r* from the wire is:

**$B = (\mu_0 I) / (2 \times \pi r)$**, written in words as B equals mu-zero times I divided by two pi times r.

Here $\mu_0$ is the permeability of free space, equal to $4 \pi \times 10^{-7}$ T m/A. The field becomes weaker as the distance from the wire increases.

### 3.2 Circular loop

At the centre of a circular loop of radius *R* with *N* turns, the field is $B = \mu_0 N I / (2R)$. The field lines pass through the loop in one direction, so one face of the loop behaves like a north pole and the other face like a south pole.

### 3.3 Solenoid

A **solenoid** is a long coil of wire wound in the shape of a helix. Inside a long solenoid the magnetic field is almost uniform and parallel to the axis. The field inside is $B = \mu_0 n I$, where *n* is the number of turns per unit length. Outside the solenoid the field is very weak. Placing a soft iron core inside the solenoid increases the field many times, and this arrangement is called an electromagnet.

| Conductor | Field formula | Key feature |
|---|---|---|
| Long straight wire | $B = \mu_0 I / (2 \pi r)$ | Circular field lines, falls with distance |
| Circular loop (centre) | $B = \mu_0 N I / (2R)$ | Acts like a short bar magnet |
| Long solenoid (inside) | $B = \mu_0 n I$ | Uniform field, strong inside, weak outside |

## 4. Fundamental Laws

### 4.1 Biot-Savart law

The Biot-Savart law gives the magnetic field produced by a small element of current. The field is directly proportional to the current and the length of the element, and inversely proportional to the square of the distance from the element. The direction is perpendicular to both the current element and the line joining it to the point. This law is used to calculate fields for wires of different shapes.

### 4.2 Ampere's circuital law

Ampere's law states that the line integral of the magnetic field around any closed loop equals $\mu_0$ times the total current passing through the loop. It is especially useful for situations with high symmetry, such as long straight wires, solenoids and toroids, because it allows the field to be found without difficult integration.

### 4.3 Gauss's law for magnetism

Gauss's law for magnetism states that the total magnetic flux through any closed surface is zero. This is a mathematical way of saying that magnetic monopoles do not exist and field lines always form closed loops.

## 5. Force in a Magnetic Field

### 5.1 Force on a moving charge

A charge *q* moving with velocity *v* in a magnetic field *B* experiences the **Lorentz force**. Its magnitude is F = q v B sin(theta), where theta is the angle between the velocity and the field. The force is perpendicular to both the velocity and the field. Because the force is always perpendicular to the motion, it changes the direction of the particle but never its speed, so the magnetic force does no work on the charge.

- If the charge moves parallel to the field, theta is 0 and the force is zero.
- If the charge moves perpendicular to the field, the force is maximum and the particle moves in a circle.
- At any other angle the particle follows a helical (spiral) path.

For a particle moving in a circle, the radius is r = m v / (q B). A heavier or faster particle makes a larger circle, while a stronger field makes a smaller circle. This relationship is used in mass spectrometers to separate ions by mass.

### 5.2 Force on a current-carrying conductor

A straight conductor of length *L* carrying current *I* in a field *B* experiences a force F = B I L sin(theta). The direction is found using **Fleming's left-hand rule**: stretch the thumb, forefinger and middle finger of the left hand at right angles to each other. If the forefinger points along the field and the middle finger along the current, the thumb shows the direction of the force.

### 5.3 Force between parallel wires

Two long parallel wires carrying currents in the same direction attract each other. If the currents flow in opposite directions, the wires repel. This effect is used to define the ampere, the SI unit of electric current.

## 6. Magnetic Properties of Materials

Materials respond differently to an external magnetic field. They are classified into three main groups.

| Type | Behaviour in a field | Examples |
|---|---|---|
| Diamagnetic | Weakly repelled; they develop a small opposing magnetisation | Copper, water, bismuth, gold |
| Paramagnetic | Weakly attracted; atoms have permanent moments that align slightly | Aluminium, platinum, oxygen |

| Type | Behaviour in a field | Examples |
|---|---|---|
| Ferromagnetic | Strongly attracted; can be permanently magnetised below the Curie temperature | Iron, nickel, cobalt |

Ferromagnetic materials contain small regions called **domains**, inside which the magnetic moments are already aligned. In an unmagnetised piece, the domains point in random directions and cancel each other. An external field makes the domains line up, which produces strong magnetisation. When the temperature rises above the **Curie temperature**, the material loses its ferromagnetism and becomes paramagnetic. For iron the Curie temperature is about 770 degrees Celsius.

Materials can also be described as **soft** or **hard**. Soft magnetic materials, such as soft iron, are easy to magnetise and demagnetise, so they are used in transformer cores and electromagnets. Hard magnetic materials, such as steel and alnico, keep their magnetisation and are used to make permanent magnets.

## 7. Earth's Magnetic Field

The Earth behaves as if a giant bar magnet were placed near its centre, tilted by about 11 degrees from the rotation axis. The field is believed to be generated by the motion of molten iron in the outer core, a process known as the geodynamo. Near the surface, the field is described by three quantities: the **declination** (angle between magnetic north and true north), the **inclination or dip** (angle the field makes with the horizontal) and the **horizontal component** of the field.

The magnetosphere formed by this field protects life by deflecting charged particles from the solar wind. When some particles are guided toward the poles, they collide with atmospheric gases and create the glowing displays known as auroras. The Earth's magnetic poles also move slowly over time, and over geological periods the field has reversed its polarity many times.

## 8. Applications of Magnetic Fields

| Application | How the magnetic field is used |
|---|---|
| Electric motor | A current-carrying coil in a magnetic field feels a torque and rotates, converting electrical energy to mechanical energy. |
| Generator | A coil rotated in a magnetic field produces an induced voltage, converting mechanical energy to electrical energy. |
| Transformer | A changing magnetic field in an iron core transfers energy between two coils, changing the voltage level. |
| MRI scanner | Strong fields and radio waves align and flip hydrogen nuclei in the body to build detailed images. |
| Maglev train | Magnetic repulsion and attraction lift and propel the train without wheel contact. |
| Hard disk and card readers | Tiny magnetised regions store data as binary 0s and 1s. |
| Cyclotron and mass spectrometer | Magnetic force bends charged particles into circular paths to accelerate or sort them. |
| Compass | A magnetised needle aligns itself with the Earth's field for navigation. |

## 9. Worked Examples

### Example 1: Field near a straight wire

A long straight wire carries a current of 10 A. Find the magnetic field at a distance of 5 cm from the wire.

Using $B = \mu_0 I / (2 \pi r)$, with $\mu_0 = 4 \pi \times 10^{-7}$ T m/A, I = 10 A and r = 0.05 m, we get $B = (2 \times 10^{-7} \times 10) / 0.05 = 4 \times 10^{-5}$ T, which is 40 microtesla. This is similar to the strength of the Earth's field at the surface.

### Example 2: Field inside a solenoid

A solenoid has 500 turns over a length of 25 cm and carries a current of 2 A. Find the field inside it.

The turn density is n = 500 / 0.25 = 2000 turns per metre. Then $B = \mu_0 n I = 4 \pi \times 10^{-7} \times 2000 \times 2$, which is about $5.0 \times 10^{-3}$ T, or 5 millitesla.

### Example 3: Radius of a charged particle path

A proton with speed $2 \times 10^6$ m/s enters a uniform field of 0.5 T at right angles to the field. The proton mass is $1.67 \times 10^{-27}$ kg and its charge is $1.6 \times 10^{-19}$ C. Find the radius of its path.

Using $r = m v / (q B)$, $r = (1.67 \times 10^{-27} \times 2 \times 10^6) / (1.6 \times 10^{-19} \times 0.5)$, which gives r of about 0.042 m, or 4.2 cm.

## 10. Key Points Summary

- A magnetic field is a vector field produced by magnets and by moving charges; its SI unit is the tesla.
- Field lines form closed loops, and isolated magnetic poles have never been found.
- A current in a wire produces a circular field, a loop acts like a magnet and a solenoid gives a uniform internal field.
- The Biot-Savart law, Ampere's law and Gauss's law for magnetism are the core laws describing magnetic fields.
- The Lorentz force acts perpendicular to the velocity of a charge and does no work on it.
- Materials are diamagnetic, paramagnetic or ferromagnetic depending on how they respond to a field.
- Magnetic fields are essential to motors, generators, transformers, medical imaging, data storage and transport.